Loan Management Module

1. Loan Origination System

1.1 Product Configuration Engine

- **Loan Product Types**
    a) Term Loans (short-term/long-term)
    b) Revolving Credit Lines
    c) Group Lending (with joint liability)
    d) Asset Financing
    e) Payday Loans

Loan Product Configuration

Sample Term Loan Product (TL-365):

- **Loan Amount Range:**
    - Minimum: 50,000 (local currency)
    - Maximum: 5,000,000 (local currency)
- **Repayment Tenure Options:**
    - Available in multiple durations: 30, 60, 90, 180, or 365 days
- **Interest Calculation Options:**
    - **Flat Rate Interest:** Simple interest charged on original principal
    - **Reducing Balance Interest:** Interest charged only on outstanding amount
- **Fee Structure:**
    - **Processing Fee:** 3% of loan amount or 5,000 (whichever is higher)
    - **Late Payment Penalty:** 5% per month on overdue amounts

2. Interest Calculation Methods

A) Flat Rate Interest Calculation:

- Interest is calculated on the full principal amount for the entire loan period
- Formula Example:

Total Interest = Loan Amount × (Annual Interest Rate/100) × (Loan Term in Days/365)

- Best for: Short-term loans where simple pricing is preferred

B) Reducing Balance Interest Calculation:

- Interest is calculated only on the remaining principal amount
- Formula Example:

Monthly Interest = Current Outstanding Principal × (Monthly Interest Rate/100)

- Best for: Longer-term loans where customers benefit from decreasing interest

Key Differences:

Feature	Flat Rate	Reducing Balance
Interest Basis	Original principal	Outstanding balance
Customer Cost	Higher overall	Lower overall
Complexity	Simple to calculate	Requires amortization
Common Use	Payday loans	Mortgages, term loans

Implementation Notes:

- The system automatically applies the correct formula based on product configuration
- Early repayment savings are automatically calculated for reducing balance loans
- All calculations comply with local financial regulations
    - **Special Schemes**
        - Graduated pricing (risk-based)
        - Promotional teaser rates
- **Penalty Algorithms**
    a) Late payment (daily compounding)
    b) Early repayment discounts
    c) Holiday/grace period logic

2. Disbursement & Collections

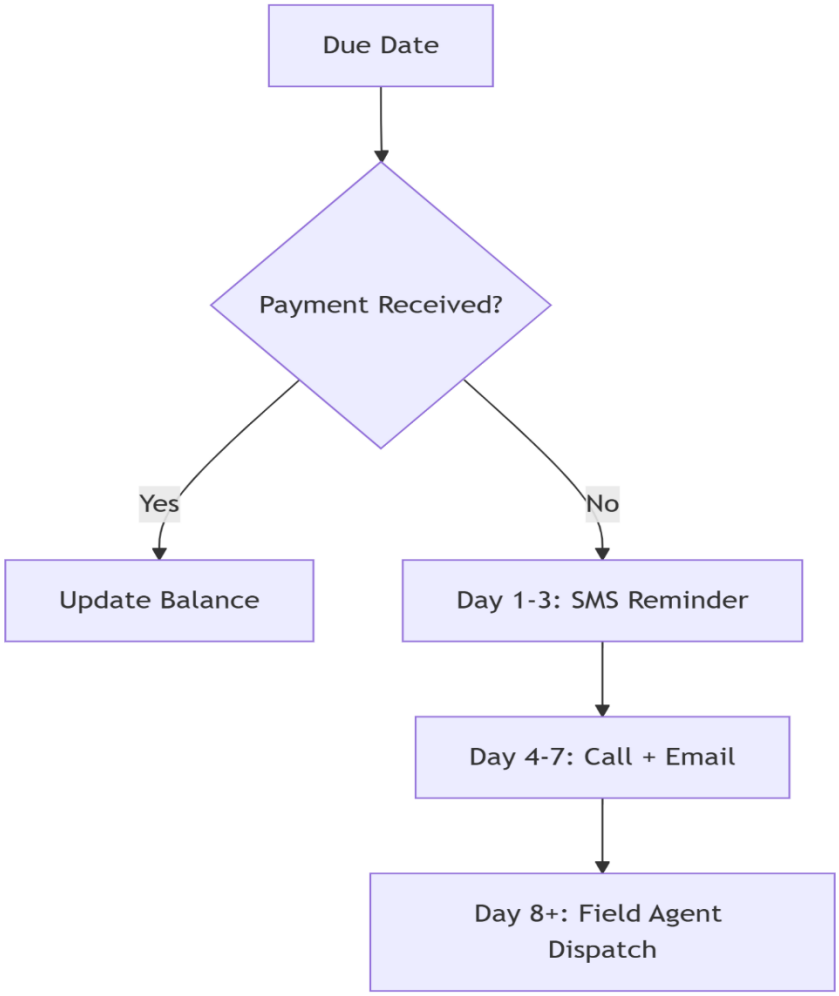
2.1 Multi-Channel Disbursement

Channel	Processing Time	Limits	Compliance Checks
Bank Transfer	2-24 hours	No max	KYC Verified
Mobile Money	Instant	5M/day	Transaction PIN
Cash Voucher	15 min	500K	OTP Verification
Cardless ATM	5 min	300K	Biometric Auth

2.2 Automated Collections

- **Payment Scheduling**
  - Fixed installment dates
  - Flexible "pay-when-you-can" options
  - Seasonal repayment calendars (harvest cycles)

Delinquency Management



- **Collection Strategies**
  - Soft collections (in-app notifications)
  - Hard collections (legal notices)
  - Debt restructuring workflows

3. Workflow Automation

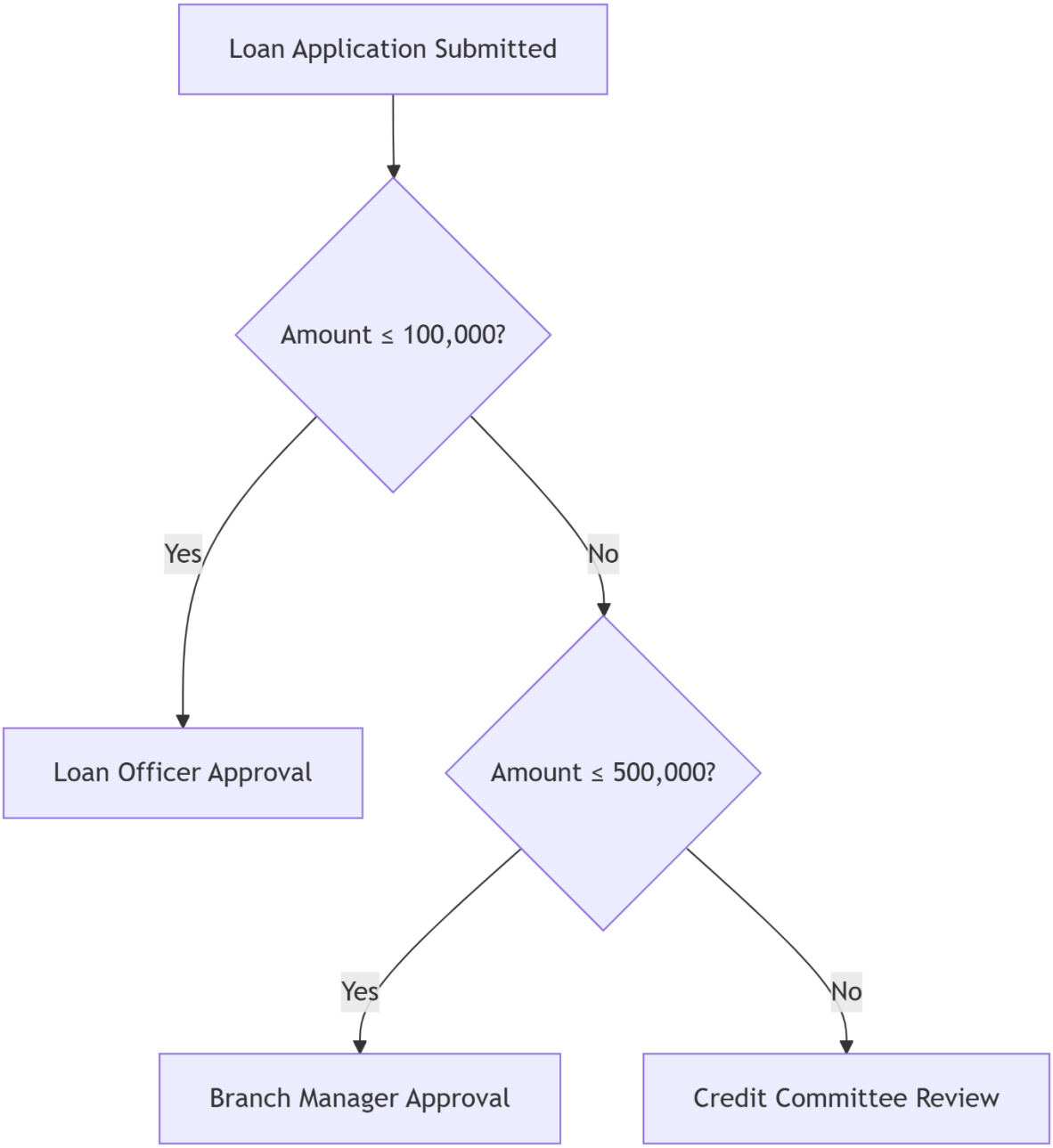
Loan Approval Hierarchy Rules

The system implements a tiered authorization system where different approval levels are required based on loan amount thresholds:

1. **Small Loans (0 - 100,000 Currency Units)**
    - Requires approval from: **Loan Officer** only
    - Typical for: Microloans, quick-disbursement products
2. **Medium Loans (100,001 - 500,000 Currency Units)**
    - Requires dual approval from:
        - **Loan Officer** (first reviewer)
        - **Branch Manager** (final approval)
    - Typical for: Individual business loans, emergency financing
3. **Large Loans (500,001+ Currency Units)**
    - Requires committee approval:
        - **Credit Committee** (3-5 members, majority vote required)
        - Includes: Senior managers, risk officers, and finance heads
    - Typical for: SME loans, mortgage facilities, corporate lending

Key Features:

- Amount thresholds are fully configurable by System Administrators
- Escalation happens automatically when application exceeds current approver's limit
- All decisions are logged with timestamps and approver comments
- Exception handling allows for urgent overrides with audit trails



- **Exception Handling**
  - Override requests
  - Committee voting system
  - Audit trail of all decisions

3.2 Digital Contracting

- **E-Signature Flow**
  1. System generates contract
  2. SMS link to borrower
  3. Biometric verification

4. Digital signing (compliant with eIDAS)
5. Auto-archiving in document management

- **Smart Contract Features**
  - Auto-triggered clauses (e.g., collateral seizure)
  - Payment-linked contract terms

4. Technical Architecture

4.1 Core Integrations

- **Payment Processors**
  - Flutterwave
  - MTN Mobile Money API
  - Bank SDKs (Equity, Stanbic)
- **Regulatory Systems**
  - Credit Reference Bureau feeds
  - Central Bank reporting module
  - Tax authority interfaces

4.2 Security Framework

- **Data Protection**
  - PCI-DSS Level 1 for payment data
  - Tokenization of sensitive fields
  - Regular HSMs key rotation
- **Fraud Prevention**
  - Velocity checks (multiple applications)
  - Geo-fencing (branch territory enforcement)
  - Device fingerprinting

Reporting & Analytics

5.1 Real-Time Monitoring

- **Loan Book Dashboard**
  - Disbursement velocity
  - Portfolio-at-risk heatmap

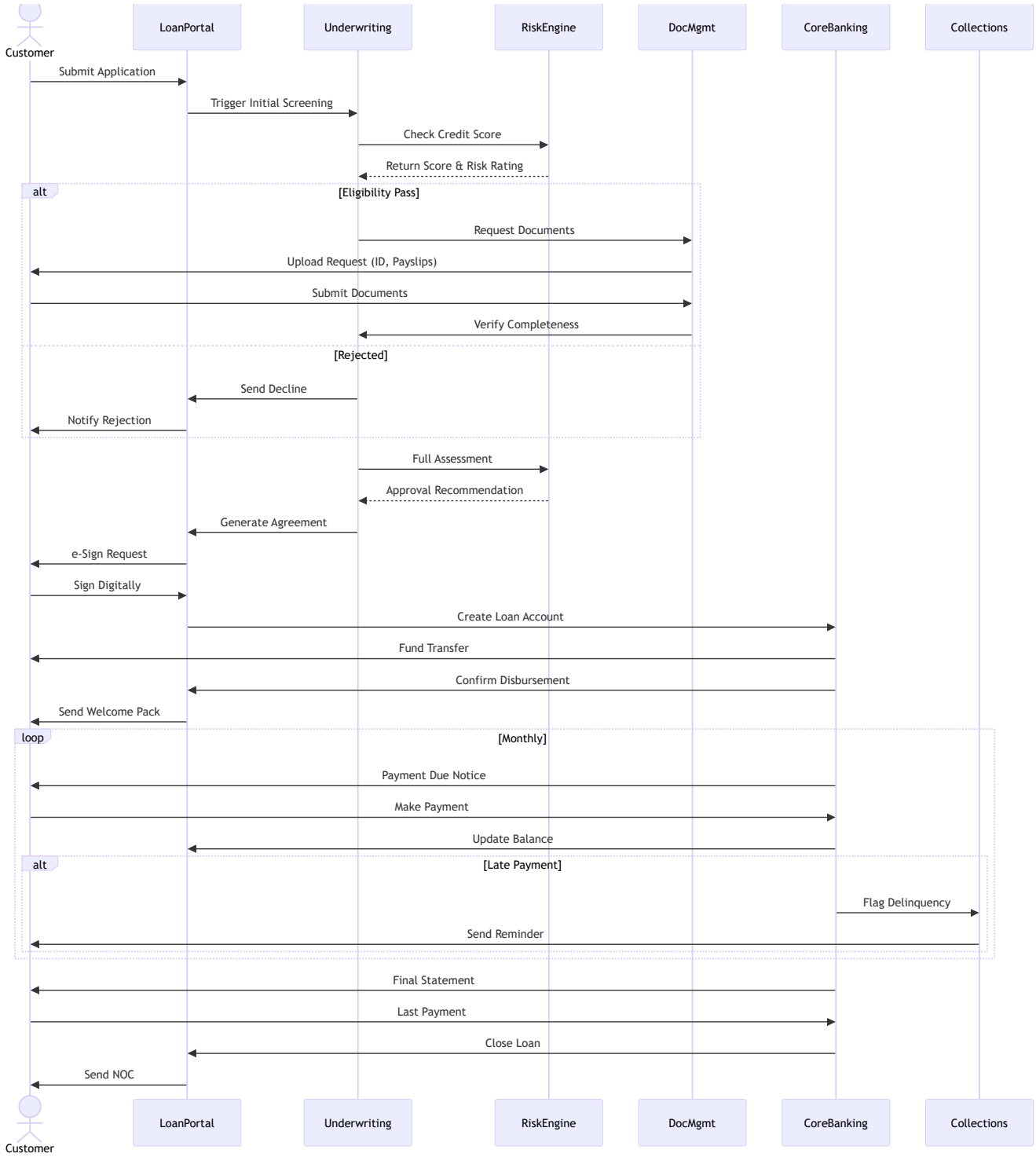
    - Yield curve analysis

5.2 Regulatory Reports

- **Automated Submissions**
    - CRB monthly files
    - Central Bank returns
    - Anti-Money Laundering (AML) alerts

Key Functions of a CRB:

1. **Credit Reporting** – Maintains records of borrowers' credit history (loans, repayments, defaults).
2. **Risk Assessment** – Provides credit scores to lenders for loan approval decisions.
3. **Fraud Prevention** – Helps detect identity theft and duplicate loan applications.
4. **Regulatory Compliance** – Ensures lenders follow financial laws (e.g., Uganda's **Tier 4 Microfinance Act**).


Customer
LoanPortal
Underwriting
RiskEngine
DocMgmt
CoreBanking
Collections
Submit Application
Trigger Initial Screening
Check Credit Score
Return Score & Risk Rating
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Request Documents
Upload Request (ID, Payslips)
Submit Documents
Verify Completeness
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Notify Rejection
Full Assessment
Approval Recommendation
Generate Agreement
e-Sign Request
Sign Digitally
Create Loan Account
Fund Transfer
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Send NOC
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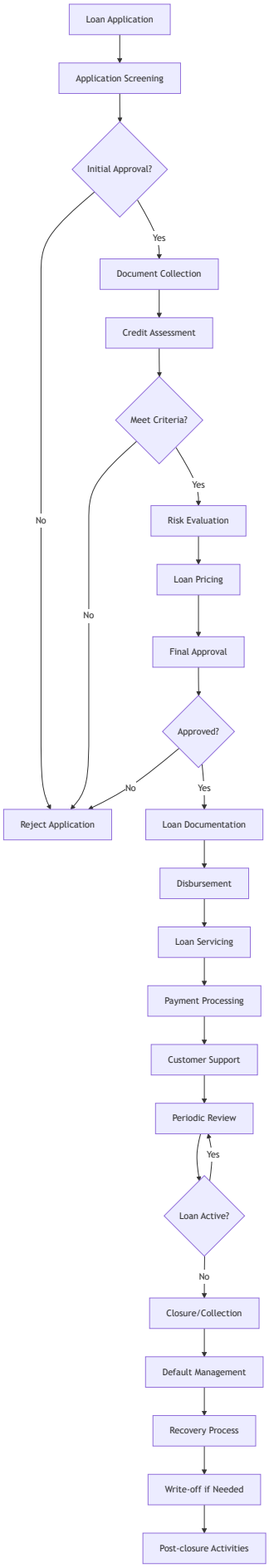
DocMgmt
CoreBanking
Collections


Loan Application
Application Screening
Initial Approval?
Yes
No
Document Collection
Credit Assessment
Meet Criteria?
Yes
No
Risk Evaluation
Loan Pricing
Final Approval
Approved?
No
Yes
Reject Application
Loan Documentation
Disbursement
Loan Servicing
Payment Processing
Customer Support
Periodic Review
Yes
Loan Active?
No
Closure/Collection
Default Management
Recovery Process
Write-off if Needed
Post-closure Activities